LiPo Battery Monitor

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Design Requirements

- We are tasked with creating a LiPo Battery Monitor that can simply monitor the battery level of a given battery to a user.
- We also want this monitor to be able to notify the user when battery voltage reaches a certain point based on user input.
- It needs to be reliable, easy to implement, and straight forward for the user.

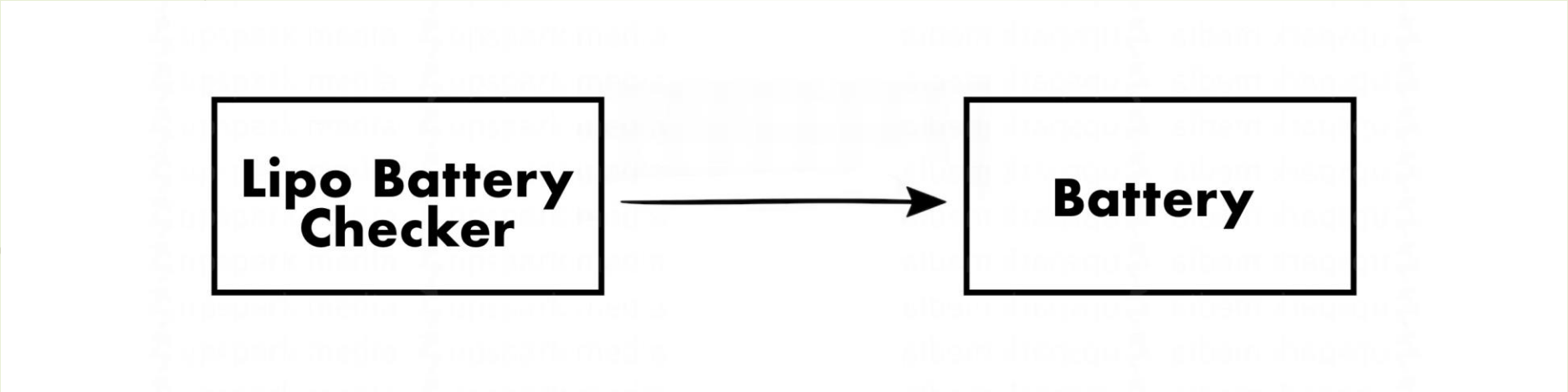
Materials and Methods

- When looking into how LiPo batteries are monitored, we found many solutions were extremely simple and cheap to build.
- After looking through some options, we decided to make use of an integrated solution available online (shown below).
- This solution is able to monitor individual cell voltage, total voltage, and notify users when voltage reaches a desired level.
- All that is needed to use it is to plug it into a battery!

BX100

Voltage Checker + Integrated Alarm

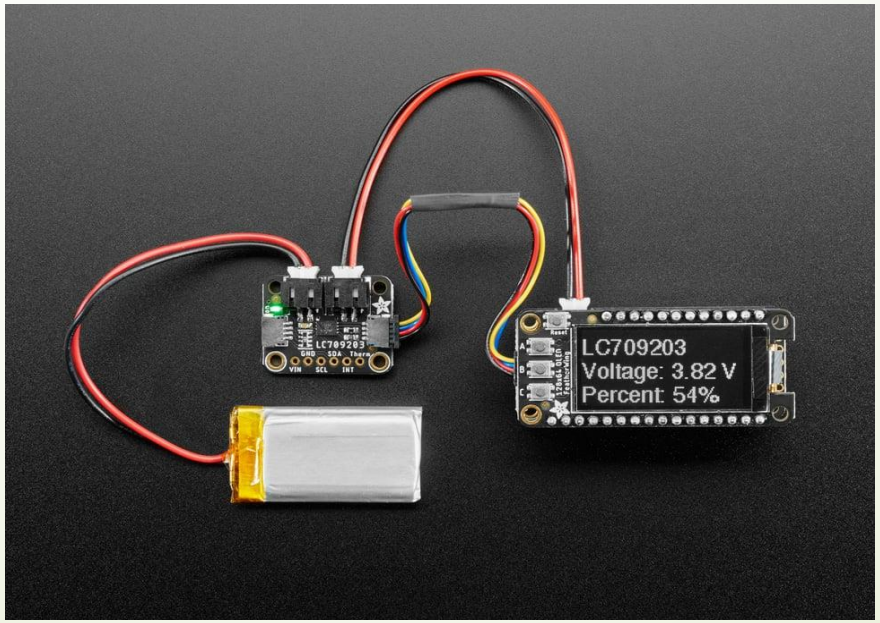
Block Diagram



HOOVO 3S 11.1V 2200mAh 50C Lipo Battery

Alternative Solutions

- We also considered making use of an Adafruit kit we found online which performs many of the same functions.
- We decided not to make use of it because integrating it into an existing system would be harder than the other solution.
- Additionally, it only supports single cell batteries, which can be a big limitation in many applications.



Testing Method

- Our team will attach the battery to the Lipo Monitoring System on a day when all teammates are available.
- The three of us will then work with a group from Goosebot's main class to maneuver the robot, in order to check the accuracy of the battery monitor.

Timeline and Roles

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
4/13 **Complete Design** **Submit Presentation** **Order Parts**	4/14 **Present Design**	4/15 **Connecting to battery**	4/16 **Testing**	4/17 **Testing**	4/18 **Demo**	4/19 **Draft Report**
4/20 **Report and Demo Due**	4/21	4/22	4/23	4/24	4/25	4/26

We all plan to complete each phase of the project together and collaborate throughout the process.

We will design, build, test, and write the report as a team.

Thank you!